

CTCH 605 Project 3
Introduction to Cybersecurity

Security Event and Incident Report Template

Prepared By: *GRAYSON GYASI*

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Introduction

The purpose of this investigation was to determine whether FICBANK experienced a cybersecurity incident and to identify potential Indicators of Compromise (IoCs) within the organization's environment. By analyzing the provided artifacts, including web server logs and network packet captures, I conducted a preliminary digital forensic investigation to identify suspicious activity, determine possible attack methods, and recommend mitigation strategies. The investigation also focused on documenting evidence and maintaining proper records that could support future incident response activities.

Objectives

1. Identify potential Indicators of Compromise (IoCs) in FICBANK's systems architecture.
2. Conduct a preliminary investigation using fundamental digital forensics techniques.
3. Document your findings using the Security Event and Incident Report Template.

Definitions

Indicator of Compromise (IoC): It is often described in the forensics world as evidence on a computer that indicates that the security of the network has been breached (Baker, 2022)

Digital Forensics: This focuses on collecting and analyzing digital evidence after a security incident (Palo Alto Networks, n.d.)

Artifact: NIST (2006) explains an artifact as data or information left behind by user activity, applications, or system processes that can be analyzed during a digital forensic investigation.

Packet Capture (PCAP): A Packet Capture (PCAP) file is a collection of recorded network communications that allows investigators to analyze network activity, identify suspicious behavior, and reconstruct events during a cybersecurity investigation (Palo Alto Networks, n.d.)

HTTP POST Request: HTTP POST request is a method used by web clients to send data to a web server for processing (Fielding & Reschke, 2022)

Web Shell: A web shell is a script or program placed on a web server that enables an attacker to execute commands remotely through a web interface (OWASP Foundation, n.d.)

Predictions

1. What types of IoCs do you predict might be found during the investigation?

I predicted that the investigation would reveal suspicious IP addresses, unauthorized access attempts, malware-related artifacts, unusual network traffic, suspicious file transfers, and evidence of reconnaissance activity.

2. What tools, techniques, and practices do you predict will be most useful?

I expected Wireshark, log analysis, and TrID analysis to be the most useful techniques. Proper evidence collection and documentation practices would also help support the investigation.

3. What do you expect the outcomes to be?

I expected to identify suspicious activity, determine possible attack methods, identify affected systems, and develop recommendations to reduce future risk.

Events and Indicators of Compromise (IoC) Work

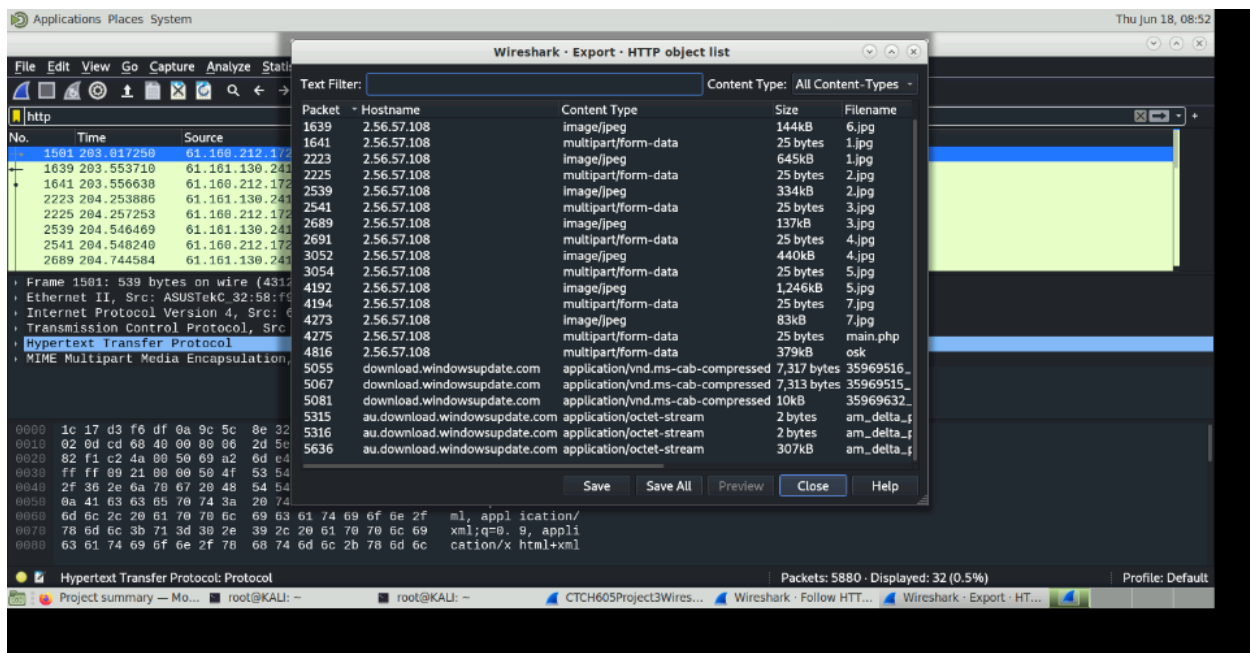
Products

Reporting Tool

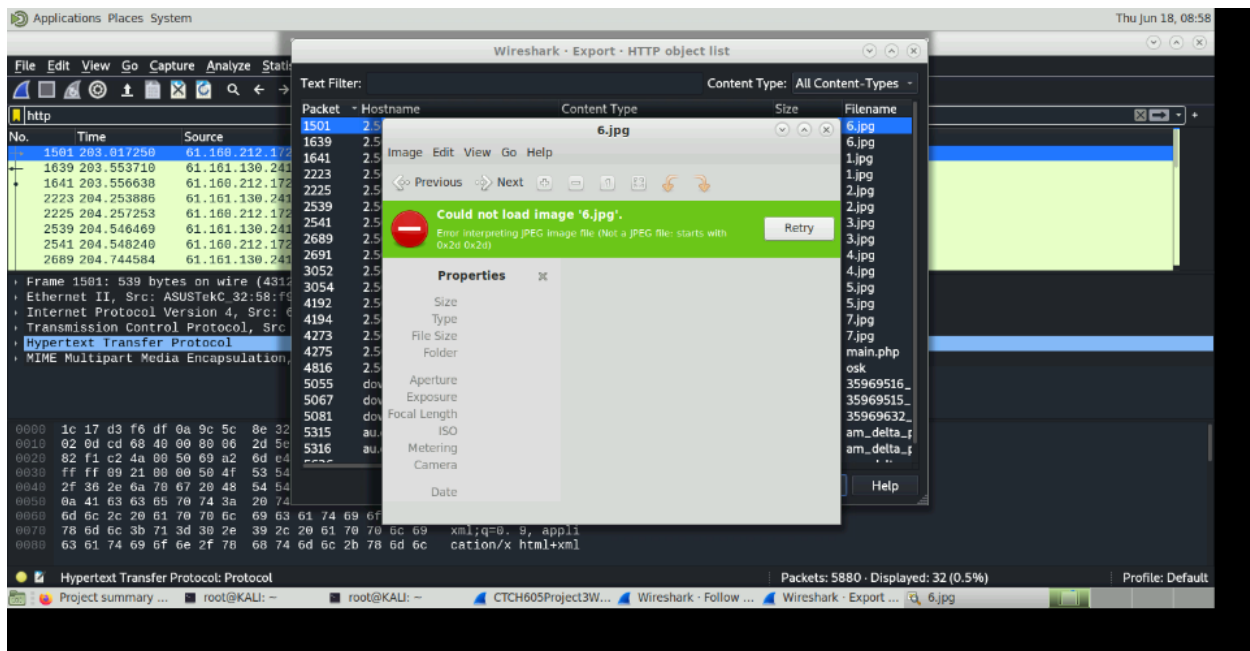
I selected Dradis as the reporting tool for this investigation. Dradis provided a centralized platform for documenting findings, storing artifacts, tracking issues, organizing methodologies, and generating reports.

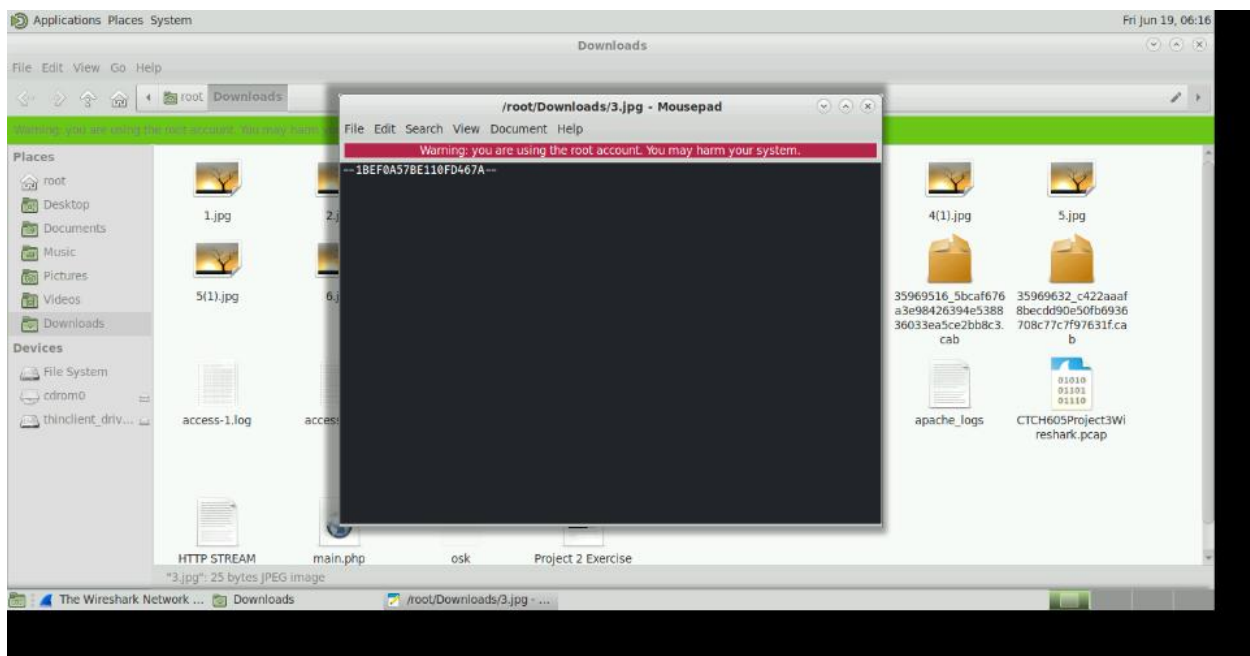
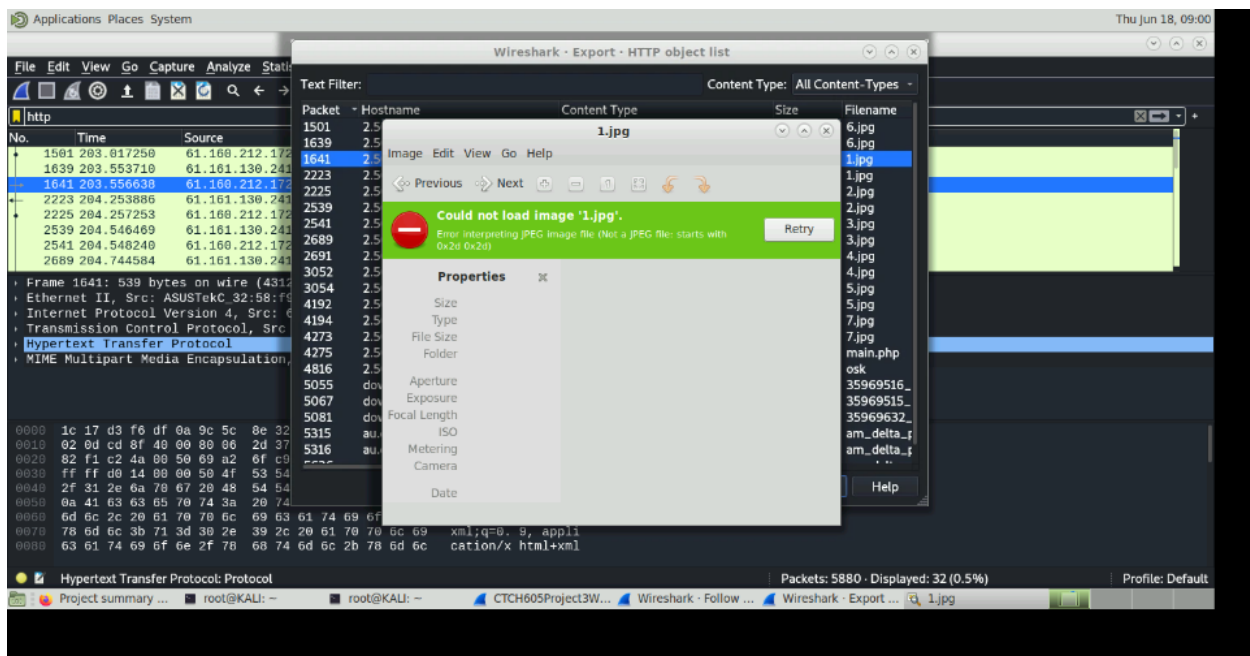
Wireshark Analysis Findings

Using Wireshark, I analyzed the provided packet capture file and filtered traffic using HTTP protocols. During the investigation, I observed multiple HTTP POST requests, file transfer activity involving several .jpg files, a file named main.php transferred between systems, communications involving external host 2.56.57.108 and multiple files transferred using multipart/form-data uploads.



When reviewing the exported HTTP objects, I discovered that several files with .jpg extensions could not be interpreted as valid JPEG images. For instance, 6.jpg, 1.jpg and 3.jpg displayed that images could not be loaded. This finding suggests that the files may have been disguised or improperly reconstructed. Additionally, the file main.php appeared suspicious because PHP files are commonly used for web shells and malicious scripts.



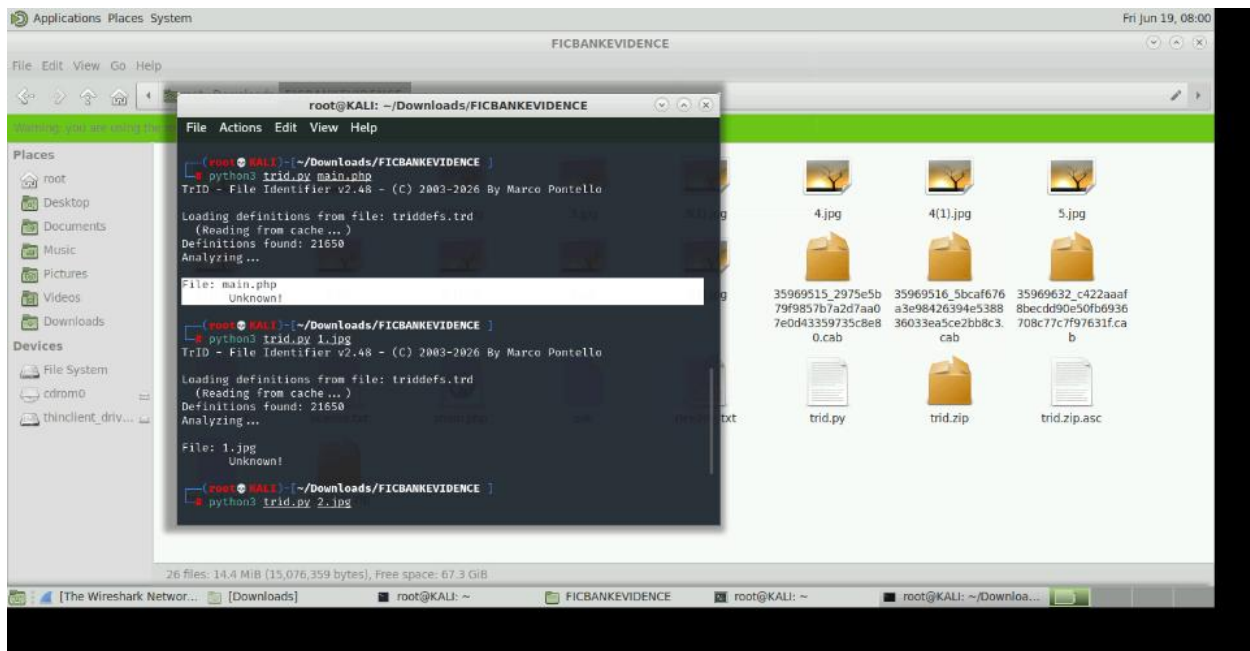


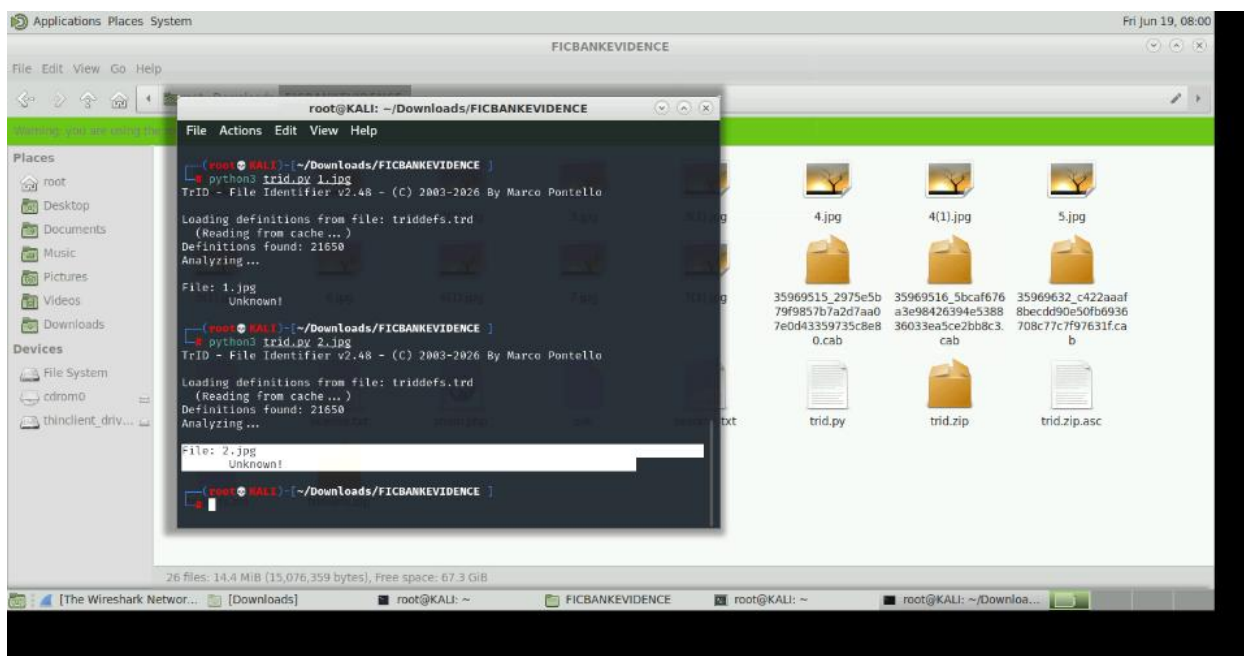
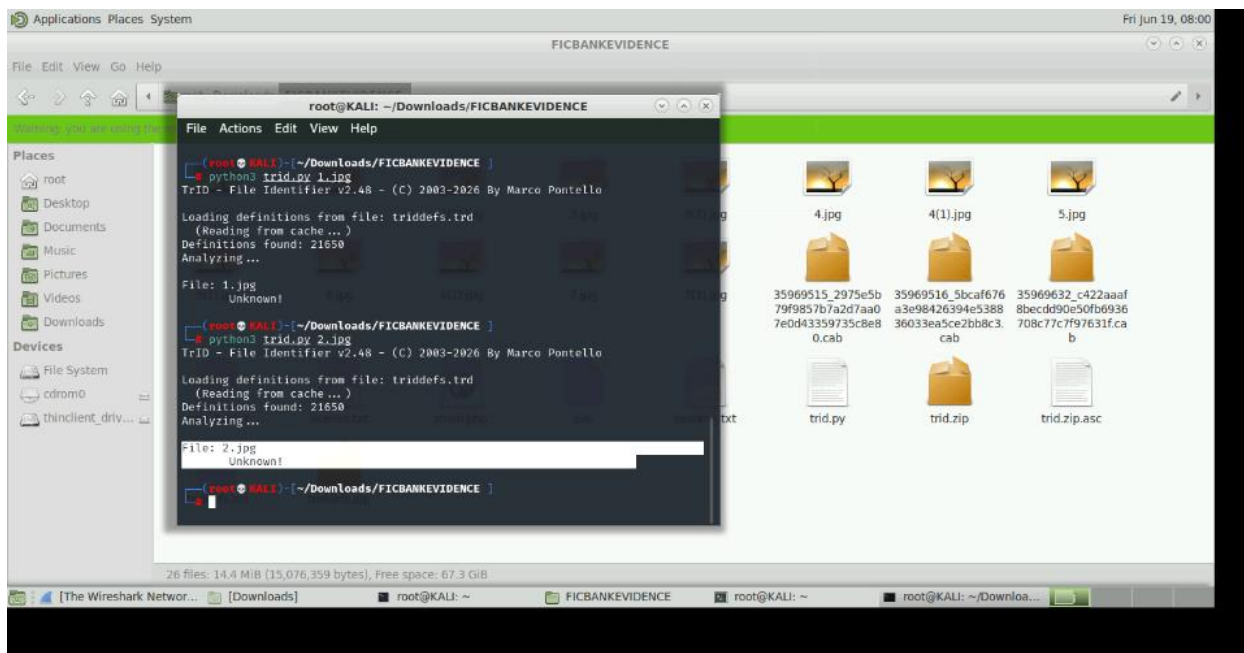
Indicators of Compromise Identified

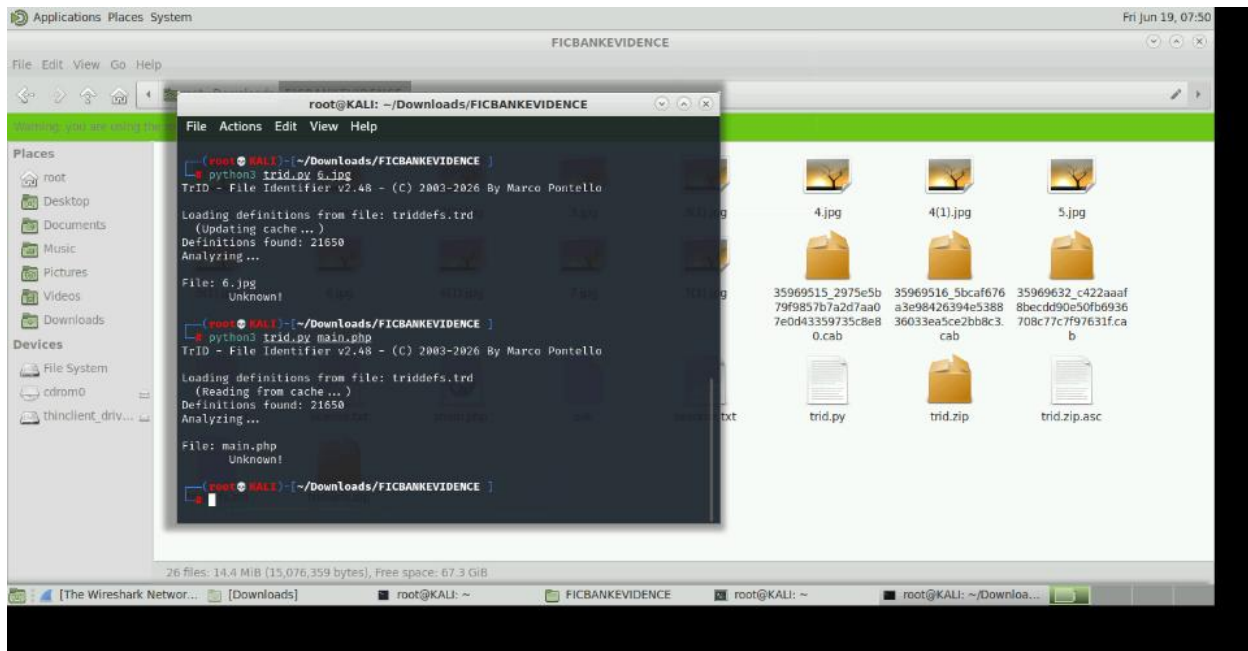
From the analysis, IOCs may not be limited to suspicious IP addresses, HTTP POST file uploads, multiple suspicious file transfers, invalid image files disguised as JPEG files and suspicious PHP file transfer.

TrID Analysis Findings

TrID was used to determine the true file types of the reconstructed files extracted from the packet capture. The files 1.jpg, 2.jpg, 6.jpg, 7.jpg and main.php were analyzed against more than 21,650 file type definitions. TrID was unable to identify either file and returned an "Unknown" result for both. These findings were significant because the files could not be validated as legitimate JPEG images or PHP scripts despite their file extensions. The results support previous observations from Wireshark, which indicated that the files did not contain valid image data and appeared to contain multipart/form-data upload information. The inability to identify the files suggests that they may have been incomplete, corrupted, or intentionally disguised, making them potential Indicators of Compromise (IoCs).





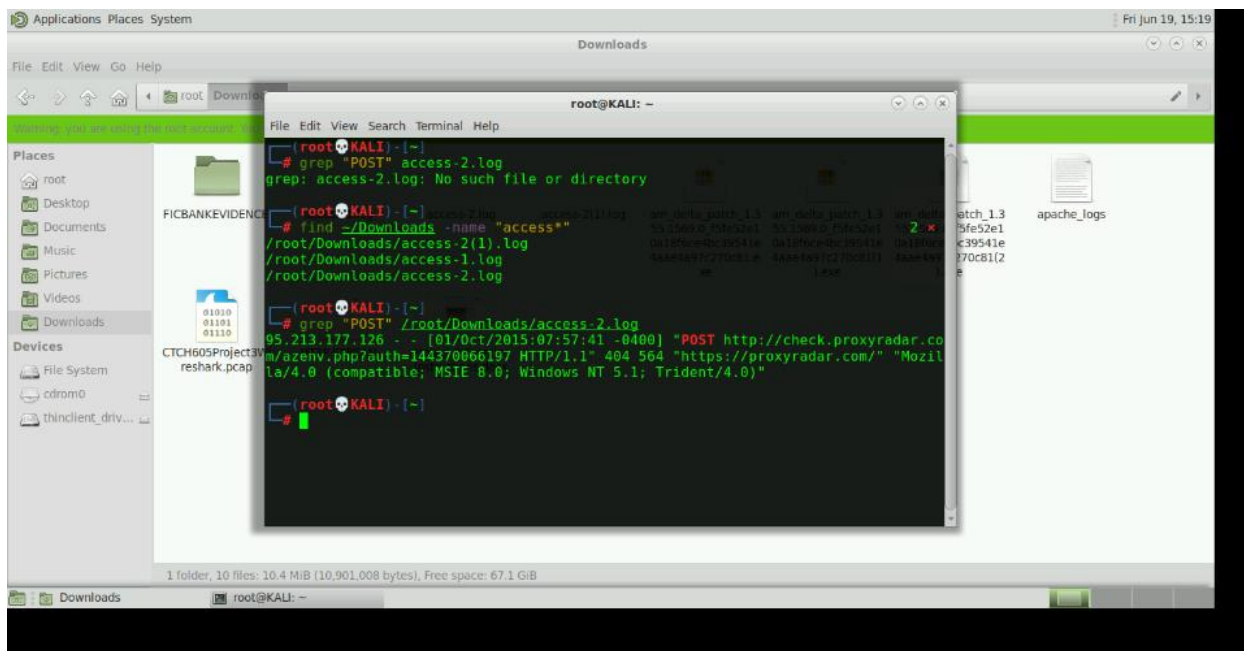


Potential Indicators of Compromise Identified

This may not be limited to multiple HTTP POST file uploads, suspicious communication with external host 2.56.57.108, files masquerading as JPEG images, unknown file types identified through TrID analysis and suspicious file upload activity involving a file named main.php.

Log Analysis Findings

Action: `grep "POST" /root/Downloads/access-2.log`



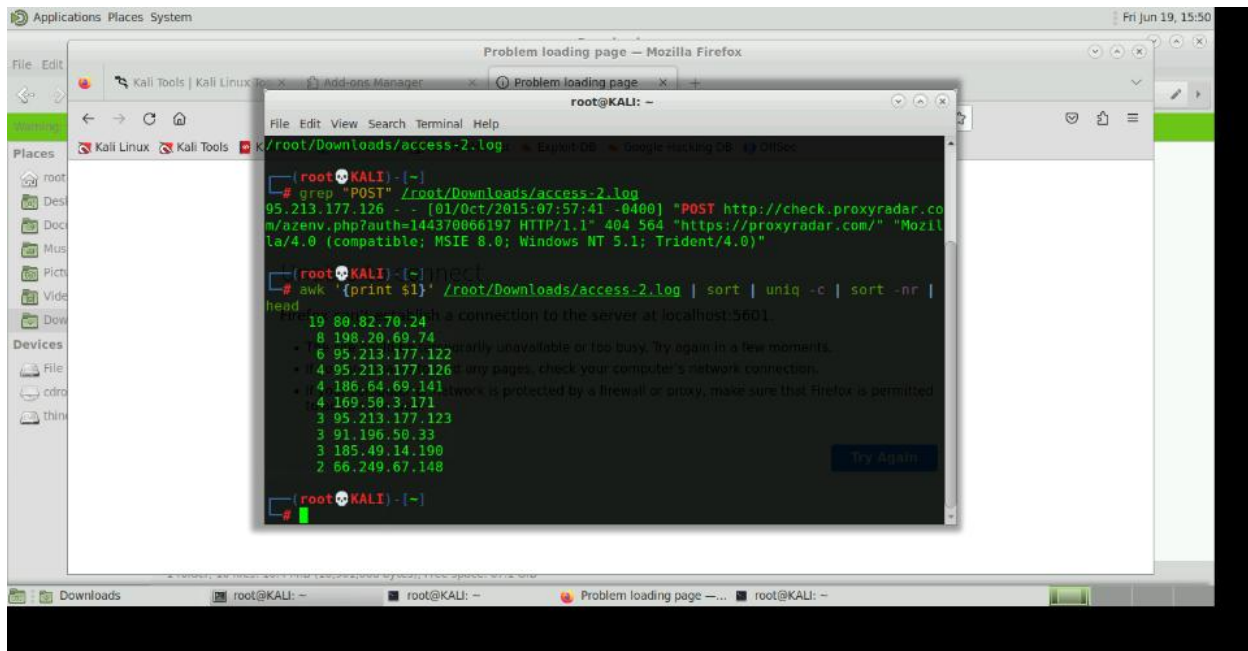
Analysis: Log analysis identified suspicious HTTP POST activity originating from the external IP address 95.213.177.126. The request targeted the web server using an HTTP POST method and referenced the domain check.proxyradar.com. The request also utilized an outdated Internet Explorer 8.0 user-agent string running on Windows XP. Although the request resulted in a 404-response code, the activity is notable because POST requests are frequently used to upload data or interact with web applications.

This finding correlates with the Wireshark analysis, which revealed multiple HTTP POST requests involving file transfer activity. The packet capture also showed the transfer of several suspicious files, including files with .jpg extensions and a file named main.php. The combination of POST requests, suspicious file transfers, and unknown file types identified through TrID analysis suggests that an attacker may have been attempting to upload or transfer files through the web application.

Most Active IP address

Action: `awk '{print $1}' /root/Downloads/access-2.log | sort | uniq -c | sort -nr | head`

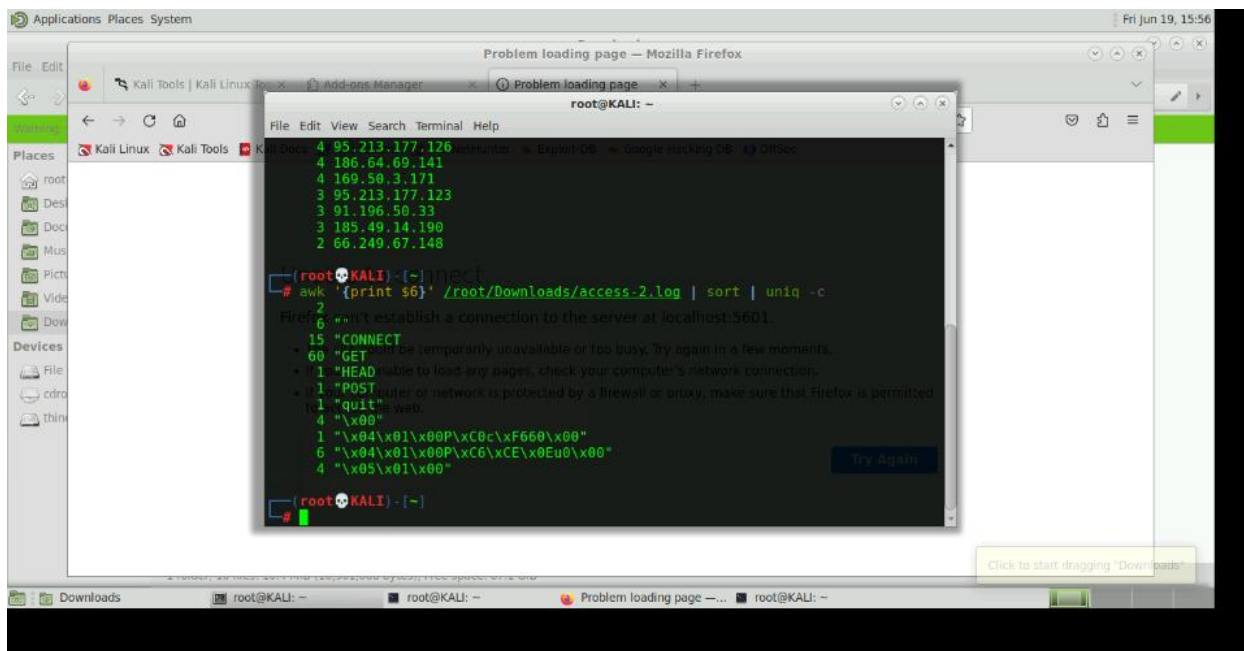
Result:



Analysis of the access-2.log file identified several external IP addresses interacting with the web server. The most active source was 80.82.70.24 with 19 requests, followed by 198.20.69.74. A POST request originating from 95.213.177.126 was observed, indicating an attempt to submit data to the web application. This finding correlates with the Wireshark packet capture, which revealed HTTP POST traffic and suspicious file upload activity. Together, the evidence suggests that external systems were interacting with the web application and may have been attempting to upload or transfer files as part of the attack.

HTTP methods found in the log

Action: `awk '{print $6}' /root/Downloads/access-2.log | sort | uniq -c`



Analysis of the HTTP methods within the access-2.log file identified 60 GET requests, 15 CONNECT requests, 1 HEAD request, 1 POST request, and 1 QUIT request. The large number of GET requests suggest reconnaissance activities. A POST request was observed originating from an external IP address and is particularly significant because POST requests are commonly used to transfer or upload data. This finding correlates with the Wireshark packet capture, which revealed HTTP POST activity involving suspicious file transfers.

Reflections

General Questions

1. What digital forensic tools did you use during the investigation?

During the investigation, I used several digital forensic and analysis tools to examine the provided artifacts. Wireshark was used to analyze the packet capture file and reconstruct HTTP objects transferred across the network. TrID was used to identify the true file types of suspicious files extracted from the packet capture. I also used native Linux tools such as grep to search and analyze the access-2.log file for suspicious activity.

2. What was your approach to identifying potential IoCs?

My approach was to analyze multiple sources of evidence and look for indicators that suggested malicious activity. I began by reviewing the packet capture in Wireshark to identify unusual network communications, file transfers, and HTTP POST requests. I then analyzed the access-2.log file to identify suspicious IP addresses, unusual HTTP methods, and abnormal web requests. I compared findings from both sources to identify patterns and correlations. Any

suspicious IP addresses, unknown file types, unusual network activity, or unauthorized file transfers were considered potential Indicators of Compromise (IoCs).

3. How did you verify the credibility of the artifacts from FICBANK's employees?

I verified the credibility of the artifacts by analyzing the original files provided for the assignment rather than relying on assumptions or secondary information. The packet capture and log files were examined directly using forensic tools, and findings were validated across multiple sources whenever possible. For example, HTTP POST activity identified in the access logs was correlated with file transfer activity observed in the packet capture. This helped increase confidence in the findings and ensured that conclusions were supported by evidence.

4. What challenges did you face during the investigation?

One challenge was determining the true nature of the files extracted from the packet capture. Several files had .jpg extensions but could not be opened as valid images, which initially made it difficult to determine whether they were legitimate files or potentially malicious artifacts. Another challenge was learning how to use unfamiliar forensic tools. I overcame these challenges by using TrID to identify file types, reviewing HTTP streams in Wireshark.

5. How did you ensure that the investigation met industry standards?

I followed fundamental digital forensic principles throughout the investigation. Original evidence was preserved, findings were documented carefully, and conclusions were based on verifiable artifacts. I maintained detailed notes, captured screenshots of important findings, and correlated evidence from multiple sources before drawing conclusions. I also followed a structured investigative process that included evidence collection, analysis, validation, and reporting, which aligns with common industry practices and digital forensic methodologies.

Conclusion

Analysis of the access-2.log file identified multiple external IP addresses communicating with the FICBANK web server. The most active source IP was 80.82.70.243, followed by several additional external hosts. A POST request originating from 95.213.177.126 was observed and referenced a proxy-related domain. HTTP method analysis identified 60 GET requests, 15 CONNECT requests, 1 HEAD request, and 1 POST request. The POST request indicates potential data submission or file upload activity.

These findings correlate with the Wireshark packet capture, which revealed HTTP POST communications and multiple file transfers involving JPG files and a file named main.php. Further analysis using TrID determined that .jpg files and main.php were unknown file types

despite their extensions. This suggests that the files may have been disguised, incomplete, or potentially malicious.

Based on the available evidence, the attack appears to have originated from external IP addresses interacting with the web application. The attacker likely conducted reconnaissance activities and attempted to transfer files using HTTP POST requests. The suspicious file transfers, proxy-related communications, and unknown file types represent potential Indicators of Compromise that warrant further investigation.

REFERENCES

Baker, K. (2022, October 4). *IOA vs IOC*. CrowdStrike. CrowdStrike IOA vs IOC Article

Palo Alto Networks. (n.d.). *What is digital forensics and incident response (DFIR)?* Palo Alto Networks Cyberpedia. Palo Alto Networks DFIR Article

National Institute of Standards and Technology. (2006). *Guide to integrating forensic techniques into incident response (NIST Special Publication 800-86)*. U.S. Department of Commerce. NIST SP 800-86 Guide

Roy T. Fielding, & Julian Reschke. (2022). *HTTP Semantics (RFC 9110)*. Internet Engineering Task Force. RFC 9110 HTTP Semantics

OWASP Foundation. (n.d.). *Web shell*. OWASP Web Security Testing Guide